

How does this argument become closer to the base case? Since the recursive calls decrement k and remove the heads from the `chars` arguments, eventually the k argument decrements to 0 or the `chars` argument becomes the blank string.

The Python code for generating combinations is in *combinations.py*:

Python

```
def getCombos(chars, k, indent=0):
    debugMsg = '.' * indent + "In getCombos('" + chars + "', " + str(k) + ")"
    print(debugMsg + ', start.')
    if k == 0:
        # BASE CASE
        print(debugMsg + " base case returns ['']")
        # If k asks for 0-combinations, return '' as the selection of
        # zero letters from chars.
        return ['']
    elif chars == '':
        # BASE CASE
        print(debugMsg + ' base case returns []')
        return [] # A blank chars has no combinations, no matter what k is.

    # RECURSIVE CASE
    combinations = []
    ❶ # First part, get the combos that include the head:
    head = chars[:1]
    tail = chars[1:]
    print(debugMsg + " part 1, get combos with head '" + head + "'")
    ❷ tailCombos = getCombos(tail, k - 1, indent + 1)
    print('.' * indent + "Adding head '" + head + "' to tail combos:")
    for tailCombo in tailCombos:
        print('.' * indent + 'New combination', head + tailCombo)
        combinations.append(head + tailCombo)

    ❸ # Second part, get the combos that don't include the head:
    print(debugMsg + " part 2, get combos without head '" + head + "'")
    ❹ combinations.extend(getCombos(tail, k, indent + 1))

    print(debugMsg + ' results are', combinations)
    return combinations

print('2-combinations of "ABC":')
print('Results:', getCombos('ABC', 2))
```
